Jayaram Reddy

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Research Interests

Robot Learning, Reinforcement Learning, Visual Representation Learning, Dynamic Scene Representations

Education

International Institute of Information Technology — Hyderabad, India
M.Sc in Computer Science and Engineering by Research - CGPA: 9.00/10 — *August 2022 - present*

Indian Institute of Technology Madras — Chennai, India
B. Tech in Mechanical Engineering and M. Tech in Data Sciences - CGPA: 8.15/10 — *August 2015 - July 2020*

Research Experience

Robotics Research Intern — May 2025 - Oct 2025
Naver Labs EUROPE (Spatial AI Team) — *Grenoble, France*

- Currently working on skill tokenization for efficient transfer and finetuning of skill models with sparse rewards.

Research Assistant — Aug 2022 - Present
Robotics research centre - (RRC - IIITH) — *Hyderabad, India*

- Proposed and developed an adversarial hypernetwork framework for efficient policy adaptation. Extended the framework to the multi-task setting by incorporating residual learning in the latent policy parameter space. *[Accepted at **AAAI 2025 PRL and GenPlan workshop**]*. Resulted in patent at TCS Research Labs India.
- Worked on Dynamic Scene representation, for the reconstruction and novel view synthesis of dynamic urban scenes which can explicitly track the moving objects using few learned anchor points which significantly reduces the number of parameters for motion estimation of dynamic objects. *[Accepted at **ICCV 2025**]*.
- Worked on diffusion based approach for motion planning which employs ensemble of costs. In addition to achieving success rates close to state-of-the-art deep learning based planners across various scenes, our approach also works well in out-of-distribution scenarios which is the main drawback of deep learning based planners. ***[ICRA 2024]***
- Worked on Self supervised registration of RGB-D videos by exploiting cycle consistency property which comes close to state-of-the-art among self supervised approaches. ***[ICRA 2024]***.

Computer Vision Intern — Jul 2017 - Feb 2018
Detect Technologies (Noctua AI) — *Chennai, India*

- Worked on detection of rusted areas and level of damage on pipelines. Level of damage (mild or severe) is found based on area of contour around the rusted area

Publications

* indicates Equal contribution

1. **Gen-HypRL: Adversarial Generative Hypernetworks for Efficient Policy Adaptation**
   Accepted at AAAI 2025 PRL Workshop and GenPlan workshop, extension under review at ICRA 2026.
   Jayaram Reddy*, Sanket Kalwar*, Brojeshwar Bhowmick, Arun Singh, K Madhava Krishna
   [Project Page]
2. **UGSDF: SDF Guided Dynamic 3D Gaussians for Urban Scenes**
   ICCV 2025
   Siddharth Tourani*, **Jayaram Reddy***, Akash Kumbar, Satyajit Tourani, Nishant Goyal, Madhava Krishna, Muhammad Haris Khan, N Dinesh Reddy
   [Project Page]
3. **EDMP: Ensemble-of-costs-guided Diffusion for Motion Planning**
   ICRA 2024, CoRL 2023 Workshop
   Kallol Saha*, Vishal Mandadi*, **Jayaram Reddy***, Ajit Srikanth, Aditya Agarwal, Bipasha Sen, Arun Singh, Madhava Krishna
   [Project Page][Paper Link]

4. **Leveraging Cycle-Consistent Anchor Points for Self-Supervised RGB-D Registration**
   ICRA 2024
   Siddharth Tourani*, **Jayaram Reddy***, Sarvesh Thakur*, K Madhava Krishna, Muhammad Haris Khan, N Dinesh Reddy
   [Paper Link]

Work Experience

Quantitative Developer June 2020 - Jan 2022
Australia and New Zealand Banking Group - ANZ *Bangalore, India*

- Worked closely with Rates team in Singapore on LIBOR Transition and built THOR curve which is replacement to LIBOR in THAI currency
- Implemented fallback curves in non-USD LIBOR currencies and also optimizing the code and fixing production issues in initial days of the work.
- Mentored new joining Quant developers, providing guidance in setting up and understanding pricing, risk calculation of the trades in the ANZ's SKY codebase

Quantitative Research Intern May 2019 - July 2019
JPMorgan Chase - JPMC *Mumbai, India*

- Worked on project which is used for suggesting the optimal strategy to the clients while trading using ensemble regression techniques.
- Integrated the tool with Client Clustering where clients are mainly clustered into two categories namely Quantitative and Institutional for increasing performance of tool

Engineering Intern Dec 2017 - Feb 2018
Caterpillar *Chennai, India*

- Created an optimization tool called as beam calculator which optimizes cross section beam without failing subjected to given loading conditions
- The tool allows user to select the material, give the loading conditions and choose between various geometries as well as various dimensions for the same cross section

Awards and Honours

- Reliance Foundation Postgraduate Scholarship *(May 2023)*
  (Awarded to only 100 PG students across the country every year.)
- Secured *AIR 119* in IIT JEE Mains (2015). **Among the top 0.092% of over 1.5 million participants**.
- Secured *AIR 1384* in IIT JEE Advanced (2015). **Among the top 0.1% of over 125,000 qualified participants**.
- **Awarded INR 200,000 funding** for a team project displayed at Envisage v5.0, Shaastra 2016, IIT Madras.

Selected Projects

Robot Joint Pose estimation - RoPoEs Aug 2022 - Dec 2022
[Project Code][Project Report]

- Worked on estimating the robot pose and joint angles using 3D Vision and Pose estimation techniques
- Cameras in the scene capture the manipulator and an hourglass network is trained to predict the 2D keypoint locations for each joint in the manipulator. These are called as belief maps. Detected 2D keypoints are then sent to triangulation module for estimating 3D joint positions and joint angles are calculated
- Automated data generation for training hourglass network by installing cameras in various circular tracks around the arm, with their view aimed at the arm in the center in **PyDrake** simulator.

Video Captioning Jan 2019 - May 2019

- Developed a basic Video captioning system using Action VLAD as encoder and a single hidden layer LSTM based RNN as decoder

Persistence of Vision - POV Jun 2016 - Jan 2017
[Project Showcase][Project Report]

- Worked in a team of 5 talented peers with the aim of creating a video display with single arm of LEDs rotating at very high rpm. Its an optical illusion whereby multiple discrete images blend into a single image in the human mind, and makes us believe that these multiple images are part of a single continuous display

- Worked on video processing to streamline data for a set of PCBs and played a significant role in assembling and efficiently designing the rotating arm. This effort led to a notable improvement in speed, increasing it from 250 rpm to 540 rpm, creating a visually appealing display
- This project was part of the Techno-Cultural Show, **Envisage v5.0, Shaastra 2016** of IIT MADRAS and was displayed in front of huge crowd of almost 3000 people. **Our project was funded INR 200000**.

Teaching Experience

Head Teaching Assistant Aug 2024 - Nov 2024
Graduate-Level Mobile Robotics Course *IIIT Hyderabad, India*

- Designed two course assignments, mid-semester exam and developed project ideas for 14 teams in collaboration with two other TAs.
- Guided five teams on projects such as Mesh Reconstruction, Camera Pose Estimation, Stereo Visual Odometry, and Semi-Global Matching for dense reconstruction.

Head Teaching Assistant Jan 2025 - May 2025
Topics in Reinforcement Learning *IIIT Hyderabad, India*

- Designed course assignments and conducted few classes on basic Deep RL algorithms and Policy Gradients.

Instructor at RRC summer school
RRC Summer School (2023, 2024, 2025) *IIIT Hyderabad, India*

- Conducted a session on Diffusion models for Robotics in 2025 summer school, covering topics of guidance in diffusion, motion planning, World models.
- Conducted a session on Reinforcement Learning in 2024 summer school, covering topics of TD-Learning, Policy Gradients, and Actor-Critic Methods.
- Conducted two sessions on feature matching, Homography and Bundle Adjustment in 2023 summer school

Responsibilities

Manager of RRC Lab's Website Jan 2023 - Present
[RRC Website]

- Responsibilities include creating event pages, updating content and adding features.

Organizer and Instructor May 2023 - June 2023
[RRC 2023 Summer school] *IIIT Hyderabad, India*

- Co-organized the annual RRC Summer School 2023 with a research partner, designing a schedule to cover essential topics for new undergraduate researchers with special focus on SLAM, Odometry and Multi View Geometry.

Technical Skills

Languages: Python, C++, C, Visual Basic
Simulation Softwares: PyDrake, PyBullet, MuJoCo, IsaacSim
Developer Tools: Git, Microsoft SQL Server
Libraries: Pytorch, Keras

References

Below are individuals who can vouch for my skills, expertise, and contributions:

- **Prof. K Madhava Krishna *(Preferred)*** - Professor, Robotics Research Center, IIIT Hyderabad - mkrishna@iiit.ac.in
- **Prof. Arun Kumar Singh** - Head, Autonomous and Kooperative Systems Lab, University of Tartu - arun.singh@ut.ee
- **Dr. Seungsu Kim** - Senior Scientist, Naver Labs Europe - seungsu.kim@naverlabs.com
- **Dr. Brojeshwar Bhowmick** - Principal Scientist, TCS Research, Kolkata, India - b.bhowmick@tcs.com

Demonstration of English Proficiency:

My entire primary and secondary school education as well as university education has been conducted in English, providing me with a strong foundation in both written and verbal communication. **All my coursework, research projects, and assignments were carried out in English**, ensuring my proficiency in the language. **Additionally, I have presented my research at international conferences, such as ICRA**, where I effectively communicated my work to a global audience. These experiences have further honed my ability to engage in academic discussions and collaborate with various researchers.